Math 110 Midterm 2 Review

Solutions

§2 Techniques

T1.

(a) If $Tv = \lambda v$, $v \neq 0$, then $0 = (T^3 - T)v = (\lambda^3 - \lambda)v$, so $\lambda \in \{0, 1, -1\}$. Since $T^3 - T = T(T - I)(T + I) = 0$ on $V \neq \{0\}$, one of the three factors is not injective, and that gives an eigenvalue.

(b) If $T - aI$ were injective, then $(T - aI)(T - bI)v = 0$ would give $(T - bI)v = 0$ for every v, so $T = bI$, which is excluded. So $T - aI$ is not injective: a is an eigenvalue. The factors commute, so the same argument gives b.

(c) Eigenvalues are roots of $x^4 + x^2 = x^2(x^2 + 1)$: $0, i, -i$. On $\mathbb{C}^2$ there are at most 2 eigenvalues, so not all three. Each pair occurs: $\operatorname{diag}(0, i)$, $\operatorname{diag}(0, -i)$, $\operatorname{diag}(i, -i)$ satisfy $T^4 = -T^2$, since $\mu^4 = -\mu^2$ for $\mu \in \{0, \pm i\}$.

(d) False. On $V = \{0\}$ every polynomial f has $f(T) = 0$ (the only vector is 0), and there is no eigenvalue. Even with $V \neq \{0\}$: rotation R on $\mathbb{R}^2$ satisfies $f(R) = 0$ for $f = x(x^2 + 1)$ (odd degree), since $R^2 + I = 0$, but R has no real eigenvalue. T1(2) needs f to split into linear factors.

T2.

(a) $T^2 - T - 2I = (T - 2I)(T + I) = 0$, distinct roots, so T is diagonalizable. Lagrange: $p_2 = \frac{x+1}{3}$, $p_{-1} = \frac{x-2}{-3}$, and $p_2 + p_{-1} = 1$. So

$$v = \tfrac{1}{3}(T + I)v - \tfrac{1}{3}(T - 2I)v,$$

with $(T - 2I)\frac{1}{3}(T + I)v = 0$ and $(T + I)\frac{1}{3}(T - 2I)v = 0$.

(b) Roots $0, 1, -1$: $p_0 = \frac{(x-1)(x+1)}{(0-1)(0+1)} = 1 - x^2$, $p_1 = \frac{x(x+1)}{2}$, $p_{-1} = \frac{x(x-1)}{2}$. So $v = (v - T^2v) + \frac{1}{2}(T^2v + Tv) + \frac{1}{2}(T^2v - Tv)$, exactly Practice Midterm A Q2.

(c) m_T divides $x^n - 1$, whose roots $e^{2\pi ik/n}$ are distinct, so T2 applies. Over $\mathbb{R}$: rotation R by $90°$ has $R^4 = I$ and no real eigenvalue.

(d) $p(T|_U) = p(T)|_U = 0$, and p has distinct roots in $\mathbb{F}$, so T2 applies to $T|_U$.

T3.

(a) Each $E(\lambda_i, T)$ is 1-dimensional (three distinct eigenvalues in dimension 3) and S-invariant, so S maps the eigenvector v_i to a multiple of itself.

(b) $S^2 - I = (S - I)(S + I) = 0$ has distinct roots, so $V = E(1, S) \oplus E(-1, S)$ (T2). Each is T-invariant (T3). In a basis that lists a basis of $E(1, S)$ then one of $E(-1, S)$, $\mathcal{M}(T)$ is block diagonal.

(c) $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) = I + N$; commuting with it is commuting with $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. For $X = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$: $XN = \left(\begin{smallmatrix} 0 & a \\ 0 & c \end{smallmatrix}\right)$, $NX = \left(\begin{smallmatrix} c & d \\ 0 & 0 \end{smallmatrix}\right)$. Equal iff $c = 0$ and $a = d$.

T4.

(a) $T\colon \mathbb{R}^{2,2} \to \mathbb{R}$, $T(A) = a_{12} - a_{21}$, is linear and surjective ($T(E_{12}) = 1$, where E_{12} has a 1 in entry $(1, 2)$ and zeros elsewhere). $\operatorname{null} T = \{A : a_{12} = a_{21}\} = U$. So $A + U \mapsto a_{12} - a_{21}$ is an isomorphism $V/U \to \mathbb{R}$, and $E_{12} + U$ is a basis of V/U (it maps to the basis 1 of $\mathbb{R}$). Check: $\dim V/U = 4 - 3 = 1$.

(b) $T(x, y, z) = (x - y,\ y - z)$ is linear with $\operatorname{null} T = \{x = y = z\} = U$, so $\operatorname{rank} T = 3 - 1 = 2$ and T is surjective. The isomorphism is $(x, y, z) + U \mapsto (x - y, y - z)$. Since $Te_1 = (1, 0)$ and $Te_3 = (0, -1)$ form a basis of $\mathbb{R}^2$, $e_1 + U,\ e_3 + U$ is a basis of $\mathbb{R}^3/U$.

(c) (i) Not well defined: $(0, 0, 0) + U = (1, 1, 1) + U$, but the rule gives 0 and 1. (ii) Well defined: $x + y - 2z$ is linear and is 0 at $(1, 1, 1)$ (T4(2)). (iii) Well defined, because adding $t(1, 1, 1)$ does not change $x - y$; but it is not linear, so it is not a linear map.

(d) $\dim \operatorname{range} T = \dim V - \dim \operatorname{null} T \le \dim V - \dim U$ because $U \subseteq \operatorname{null} T$, with equality iff $\dim \operatorname{null} T = \dim U$, iff $\operatorname{null} T = U$. In quotient language: T induces $\tilde{T}\colon V/U \to W$ with the same range (Prop. 42), and $\tilde{T}$ is injective iff $\operatorname{null} T = U$.

T5.

(a) $T|_U$ is injective (restriction of injective), so it is invertible on U (finite dimension): $T(U) = U$. If $Tv \in U$, then $Tv = Tu$ for some $u \in U$, and injectivity gives $v = u \in U$.

(b) $T = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $U = \operatorname{span}(e_1)$: $T|_U = 0$ and $T/U = 0$ are diagonalizable, T is not. No contradiction: (1) says m_T divides $m_{T|_U} m_{T/U} = x \cdot x$, and indeed $m_T = x^2$, which has a repeated root.

(c) Apply $\prod_{i<k}(T - \lambda_i I)$ to $\sum_j v_j \in U$ (invariance keeps it in U): all terms with $j < k$ die, leaving $\prod_{i<k}(\lambda_k - \lambda_i)v_k \in U$, so $v_k \in U$. For any other j, apply $\prod_{i\neq j}(T - \lambda_i I)$ instead: every term except v_j dies, leaving $\prod_{i\neq j}(\lambda_j - \lambda_i)v_j \in U$, so $v_j \in U$.

T6.

(a) If $\sum_{j<m} a_j T^j v = 0$ with not all $a_j = 0$, let i be the smallest index with $a_i \neq 0$, and apply T^{m-1-i}: every term with $j > i$ becomes $T^{m-1-i+j}v = 0$, leaving $a_i T^{m-1} v = 0$, so $a_i = 0$: contradiction.

(b) Let v be cyclic and $v = \sum w_i$ with $w_i \in E(\lambda_i, T)$, λ_i distinct, m of them. Then $T^j v = \sum \lambda_i^j w_i \in \operatorname{span}(w_1, \dots, w_m)$, so $\dim V = \dim \operatorname{span}(v, Tv, \dots) \le m$. And $m \le \dim V$ always.

(c) $v = (1,1,1)$: v, Tv, T^2v have coordinates $(1,1,1), (1,2,3), (1,4,9)$, independent (row reduce the 3×3 matrix with these rows: it has rank 3). For I_2: $Iv = v$, so $\operatorname{span}(v, Iv, \dots)$ has dimension $1 < 2$.

T7.

(a) U has basis $x - 1,\ x^2 - 1$ and W has basis $x + 1,\ x^2 - 1$ (each is the null space of a nonzero functional on $\mathcal{P}_2(\mathbb{R})$, so has dimension 2). $p \in U \cap W$ iff $p(1) = p(-1) = 0$ iff $p = c(x-1)(x+1)$, so $U \cap W = \operatorname{span}(x^2 - 1)$. $\operatorname{null} \Gamma \cong U \cap W$ has dimension 1, so $\dim(U + W) = \dim(U \times W) - 1 = 4 - 1 = 3$, and $U + W = \mathcal{P}_2(\mathbb{R})$.

(b) Γ maps onto the sum, so by rank–nullity $\dim(U_1 + \cdots + U_m) = \sum \dim U_i - \dim \operatorname{null} \Gamma$. The sum is direct iff Γ is injective iff $\dim \operatorname{null} \Gamma = 0$ iff the dimensions are equal.

(c) $(0, T0) = (0,0) \in G$; $(v, Tv) + (v', Tv') = (v + v', T(v + v'))$; $\lambda(v, Tv) = (\lambda v, T(\lambda v))$. So G is a subspace. The map $v \mapsto (v, Tv)$ is linear, injective (look at the first entry) and onto G, so $\dim G = \dim V$. For the quotient, $S\colon V \times W \to W$, $S(v, w) = w - Tv$, is linear, surjective ($S(0, w) = w$), and $\operatorname{null} S = \{(v, w) : w = Tv\} = G$. By the first isomorphism theorem, $(V \times W)/G \cong W$.

(d) Send $S \in \mathcal{L}(V_1 \times V_2, W)$ to the pair (S_1, S_2) with $S_1 v_1 = S(v_1, 0)$ and $S_2 v_2 = S(0, v_2)$; this is linear in S. Its inverse sends (S_1, S_2) to the map $(v_1, v_2) \mapsto S_1 v_1 + S_2 v_2$. Check: $S(v_1, v_2) = S(v_1, 0) + S(0, v_2)$, and starting from (S_1, S_2) you get back S_1 and S_2. So the map is an isomorphism.

T8.

(a) "$\supseteq$": if $\varphi \in U^0$ and $\psi \in W^0$, then $\varphi + \psi$ is 0 on $U \cap W$. Dimensions: $\dim(U^0 + W^0) = \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0) = (n - u) + (n - w) - \dim(U + W)^0 = (n - u) + (n - w) - (n - \dim(U + W)) = n - \dim(U \cap W)$, which is $\dim(U \cap W)^0$.

(b) ($\Rightarrow$) shown in T8's why. ($\Leftarrow$) Suppose U^0 is T'-invariant but $Tu \notin U$ for some $u \in U$. By the separating functional, there is $\varphi \in U^0$ with $\varphi(Tu) = 1$. Then $(T'\varphi)(u) = 1$, so $T'\varphi \notin U^0$: contradiction.

(c) $U \neq \mathbb{F}^n$, so $U^0 \neq \{0\}$ ($\dim U^0 = n - \dim U > 0$). Any nonzero $\varphi = (c_1, \dots, c_n) \in U^0$ works.

(d) $S\colon V \to \mathbb{F}^m$, $S(v) = (\varphi_1(v), \dots, \varphi_m(v))$, has null space $\bigcap \operatorname{null} \varphi_i$ and rank $\le m$. So $\dim \bigcap \operatorname{null} \varphi_i \ge n - m$. Equality iff $\operatorname{rank} S = m$ iff S is surjective iff S' is injective; S' sends ψ_i (the ith coordinate functional on $\mathbb{F}^m$, the dual basis of $e_1, \dots, e_m$) to $\psi_i \circ S = \varphi_i$, so S' injective iff the φ_i are independent.

T9.

(a) $\chi_A = t^2 - 4t + 4 = (t-2)^2$, so $N = A - 2I = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ has $N^2 = 0$ (multiply it out). $A^k = 2^k I + k 2^{k-1} N = \left(\begin{smallmatrix} 2^k + k2^{k-1} & k2^{k-1} \\ -k2^{k-1} & 2^k - k2^{k-1} \end{smallmatrix}\right)$.

(b) $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ sends (F_n, F_{n+1}) to (F_{n+1}, F_{n+2}). Its eigenvalues are g_+, g_- (roots of $t^2 - t - 1$), with eigenvectors $(1, g_+), (1, g_-)$. $(0,1) = \frac{1}{\sqrt{5}}\big((1, g_+) - (1, g_-)\big)$, so $A^n(0,1) = \frac{1}{\sqrt{5}}\big(g_+^n(1, g_+) - g_-^n(1, g_-)\big)$; the first coordinate is F_n.

(c) By T9 with $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, the kth power is $\left(\begin{smallmatrix} \lambda^k & k\lambda^{k-1} \\ 0 & \lambda^k \end{smallmatrix}\right)$, which tends to 0 iff $|\lambda| < 1$ (then $k|\lambda|^{k-1} \to 0$ as well). Same condition as for the diagonal matrix; only the rate differs.

T10. (i) False: N. (ii) False: $N^2 = 0$ is diagonalizable, N is not. (iii) False: N with $U = \operatorname{span}(e_1)$ (T5(b)). (iv) False: rotation R on $\mathbb{R}^2$ has no invariant line. (v) False: $(1,0) + \operatorname{span}\big((-1,1)\big)$ in $\mathbb{R}^2$ does not contain 0.

§3 Drills

§3.1 Products, affine subsets and quotient spaces (Ch. 8)

1. A product space.

a) $(1,0),\ (x,0),\ (0,e_1),\ (0,e_2)$, where 0 denotes the zero of the other factor; $\dim V = 2+2 = 4$.

b) T is linear (each entry enters linearly). Injective: if $p+ax^2+bx^3 = 0$ with $\deg p \le 1$, comparing coefficients gives $a = b = 0$ and $p = 0$. Since $\dim V = 4 = \dim \mathcal{P}_3(\mathbb{R})$, an injective linear map between them is an isomorphism.

c) It is the null space of the functional $\varphi\big(p,(a,b)\big) = p(1) - a - b$, which is nonzero ($\varphi(1,(0,0)) = 1$). A nonzero functional has null space of dimension $\dim V - 1 = 3$.

2. The sum map.

a) $\Gamma\big((u_i)+(u_i')\big) = \sum(u_i+u_i') = \sum u_i + \sum u_i'$ and $\Gamma(\lambda(u_i)) = \lambda\sum u_i$.

b) Every value $u_1+\cdots+u_m$ lies in $U_1+\cdots+U_m$, and every element of $U_1+\cdots+U_m$ has this form.

c) Γ is injective iff $\operatorname{null}\Gamma = \{0\}$, i.e. iff $u_1+\cdots+u_m = 0$ with $u_i \in U_i$ forces every $u_i = 0$. That is exactly the condition for the sum to be direct.

d) Rank–nullity for Γ: $\dim(U_1+\cdots+U_m) = \dim(U_1\times\cdots\times U_m) - \dim\operatorname{null}\Gamma = \sum\dim U_i - \dim\operatorname{null}\Gamma$. So $\le$ holds, with equality iff $\operatorname{null}\Gamma = \{0\}$, iff the sum is direct by (c).

3. Affine subsets.

a) For $u \in U$: $v+u = w+\big((v-w)+u\big) \in w+U$. So $v+U \subseteq w+U$. Since $w-v = -(v-w) \in U$, the same argument gives $w+U \subseteq v+U$.

b) Pick $v+u_1 = w+u_2$ in the intersection. Then $v-w = u_2-u_1 \in U$.

c) $v \in x+W$, so $v = x+w_0$ with $w_0 \in W$. For $u \in U$, $v+u \in x+W$, so $v+u = x+w$ for some $w \in W$, and $u = w-w_0 \in W$. So $U \subseteq W$; swap the roles to get $W \subseteq U$.

d) $U = \{(x,y,z) : x+2y-z = 0\}$ and $v = (3,0,0)$: $(x,y,z) \in A$ iff $(x,y,z)-(3,0,0) \in U$.

4. Solution sets.

a) $x+y+z = 0$, $x-z = 0$: $x = z$, $y = -2z$. $\operatorname{null}T = \operatorname{span}\big((1,-2,1)\big)$.

b) $(1,1,0)$ works: $1+1+0 = 2$ and $1-0 = 1$. So the set is $(1,1,0)+\operatorname{span}\big((1,-2,1)\big)$.

c) If $Tu_0 = b$, then $Tu = b$ iff $T(u-u_0) = 0$ iff $u-u_0 \in \operatorname{null}T$ iff $u \in u_0+\operatorname{null}T$.

5. Operations on V/U.

a) $v-v' \in U$ and $w-w' \in U$, so $(v+w)-(v'+w') = (v-v')+(w-w') \in U$, so $(v+w)+U = (v'+w')+U$ (Q3(a)).

b) $\lambda v - \lambda v' = \lambda(v-v') \in U$.

c) $0+U \in W/U$ since $0 \in W$; $(w_1+U)+(w_2+U) = (w_1+w_2)+U$ and $\lambda(w+U) = \lambda w+U$ stay in W/U since W is a subspace. W/U is the quotient of the space W by its subspace U, so $\dim W/U = \dim W - \dim U$ (Prop. 40 applied to W).

d) $(1,2,5)-(1,2,-1) = (0,0,6) \in U$: equal. $(1,2,5)-(1,3,5) = (0,-1,0) \notin U$: not equal.

6. A basis of a quotient.

a) $(1,1,0,0), (0,0,1,1), e_1, e_3$ is a basis of $\mathbb{R}^4$ (it spans: $e_2 = (1,1,0,0)-e_1$, $e_4 = (0,0,1,1)-e_3$). So $e_1+U,\ e_3+U$ is a basis of V/U.

b) $(3,1,4,1) = 1\cdot(1,1,0,0)+1\cdot(0,0,1,1)+2e_1+3e_3$, so $(3,1,4,1)+U = 2(e_1+U)+3(e_3+U)$.

c) $T(x_1,x_2,x_3,x_4) = (x_1-x_2,\ x_3-x_4)$: $\operatorname{null}T = \{x_1 = x_2,\ x_3 = x_4\} = U$, and T is surjective ($Te_1 = (1,0)$, $Te_3 = (0,1)$). Isomorphism: $x+U \mapsto (x_1-x_2, x_3-x_4)$.

7. Quotients of an infinite-dimensional space.

a) $p \mapsto p(0)$ is linear, surjective onto $\mathbb{R}$, with null space U. By the first isomorphism theorem $V/U \cong \mathbb{R}$, so $\dim V/U = 1$; basis $1+U$ (it maps to 1).

b) $p \mapsto \big(p(0),p'(0)\big)$ is surjective onto $\mathbb{R}^2$ ($1 \mapsto (1,0)$, $x \mapsto (0,1)$) with null space W. So $V/W \cong \mathbb{R}^2$ and $\dim V/W = 2$.

c) $\pi\colon V \to V/\{0\}$ is surjective with null space $\{0\}$, so it is an isomorphism ($p \mapsto p+\{0\}$).

Note that V is infinite-dimensional but these quotients are finite-dimensional.

§3.2 Quotient maps and the isomorphism theorems (Ch. 9)

8. The first isomorphism theorem.

a) If $v + \operatorname{null} T = v' + \operatorname{null} T$, then $v - v' \in \operatorname{null} T$, so $Tv - Tv' = T(v - v') = 0$.
b) $\bar{T}\big((v + \operatorname{null} T) + \lambda(w + \operatorname{null} T)\big) = \bar{T}\big((v + \lambda w) + \operatorname{null} T\big) = T(v + \lambda w) = Tv + \lambda Tw$.
c) If $\bar{T}(v + \operatorname{null} T) = 0$, then $Tv = 0$, so $v \in \operatorname{null} T$ and $v + \operatorname{null} T = 0 + \operatorname{null} T$.
d) Each $Tv \in \operatorname{range} T$ equals $\bar{T}(v + \operatorname{null} T)$. So $\bar{T}\colon V/\operatorname{null} T \to \operatorname{range} T$ is an isomorphism. Differentiation D on $\mathcal{P}(\mathbb{R})$ is surjective (every polynomial has an antiderivative) with $\operatorname{null} D = \mathbb{R}$, so $\mathcal{P}(\mathbb{R})/\mathbb{R} \cong \mathcal{P}(\mathbb{R})$ via $p + \mathbb{R} \mapsto p'$.

9. Symmetric parts.

a) $\operatorname{null} T = \{A : A^T = -A\} = \operatorname{span}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. $\operatorname{range} T$ is the set of symmetric matrices: $(A + A^T)^T = A + A^T$, and a symmetric S equals $T(\frac{1}{2}S)$.
b) $A + \operatorname{null} T \mapsto A + A^T$.
c) $E_{11} + \operatorname{null} T$, $E_{22} + \operatorname{null} T$, $E_{12} + \operatorname{null} T$, where E_{ij} is the matrix with a 1 in entry (i, j) and zeros elsewhere, with images $2E_{11}$, $2E_{22}$, $E_{12} + E_{21}$ (a basis of the symmetric matrices, so the three cosets form a basis of $V/\operatorname{null} T$).
d) $\dim V/\operatorname{null} T = 4 - 1 = 3 = \dim \operatorname{range} T$.

10. Taylor coefficients.

a) $T(1) = (1, 0, 0)$, $T(x - 2) = (0, 1, 0)$, $T\big((x - 2)^2\big) = (0, 0, 2)$, so T is surjective and $\dim U = 5 - 3 = 2$. $(x - 2)^3$ and $(x - 2)^3 x$ lie in U (each has $p(2) = p'(2) = p''(2) = 0$) and are independent, so they span U: $U = \{(x - 2)^3 q : q \in \mathcal{P}_1(\mathbb{R})\}$.
b) $p + U \mapsto \big(p(2), p'(2), p''(2)\big)$.
c) Their images $(1, 0, 0), (0, 1, 0), (0, 0, 2)$ form a basis of $\mathbb{R}^3$, and an isomorphism carries bases to bases.
d) $x^4 \mapsto (16, 32, 48) = 16(1, 0, 0) + 32(0, 1, 0) + 24(0, 0, 2)$. So $x^4 + U = 16(1 + U) + 32\big((x - 2) + U\big) + 24\big((x - 2)^2 + U\big)$.

11. Maps out of a quotient.

a) Well defined: $v - v' \in U \subseteq \operatorname{null} T$ gives $Tv = Tv'$. Linear: from T. Unique: any such map has the prescribed value at every element $v + U$ of V/U.
b) $\tilde{T}(v + U) = 0$ iff $Tv = 0$ iff $v \in \operatorname{null} T$.
c) $\tilde{T}$ injective iff its null space is $\{0 + U\}$ iff every $v \in \operatorname{null} T$ lies in U iff $\operatorname{null} T \subseteq U$; together with $U \subseteq \operatorname{null} T$, iff $U = \operatorname{null} T$.
d) $\operatorname{null} T = \operatorname{span}\big((1, 1, 0), e_3\big)$, so $\operatorname{null} \tilde{T} = \operatorname{span}\big((1, 1, 0) + U\big)$, 1-dimensional. Not injective.

12. The second isomorphism theorem.

a) Linear: S is the quotient map restricted to U. Surjective: $u + w + W = u + W$ because $(u + w) - u \in W$.
b) $u + W = 0 + W$ iff $u \in W$; for $u \in U$ this means $u \in U \cap W$.
c) First isomorphism theorem applied to S. Dimensions: $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$.
d) $e_2 + e_3 \notin U$, so $U \cap W = \{0\}$ and $U + W = \mathbb{R}^3$ (dimension $2 + 1$). Both sides: $\dim U/\{0\} = 2$ and $\dim \mathbb{R}^3/W = 3 - 1 = 2$.

13. The third isomorphism theorem.

a) $0 + W \in U/W$, and sums and multiples of elements $u + W$ $(u \in U)$ are again of this form.
b) Well defined: $v - v' \in W \subseteq U$ gives $v + U = v' + U$. Linear: clear. Surjective: $v + U = S(v + W)$.
c) $S(v + W) = 0 + U$ iff $v \in U$ iff $v + W \in U/W$. The first isomorphism theorem gives $(V/W)/(U/W) \cong V/U$.
d) $\big((x, y, z) + W\big) + U/W \mapsto (x, y, z) + U \mapsto z$, using $V/U \cong \mathbb{R}$ via $(x, y, z) + U \mapsto z$.

14. A quotient operator.

a) $Ae_3 = (0, 0, 3) = 3e_3 \in U$.
b) $Te_1 = (2, 0, 1)$, so $(T/U)(e_1 + U) = 2e_1 + U$. $Te_2 = (1, 2, 1)$, so $(T/U)(e_2 + U) = e_1 + 2e_2 + U$. Matrix $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$.
c) $T|_U = 3I$: eigenvalue 3. T/U: upper-triangular, eigenvalue 2 only; it is not $2I$, so $m_{T/U} = (x - 2)^2$ and T/U is not diagonalizable. T: its eigenvalues are those of $T|_U$ together with those of T/U (Q32): 2 and 3.
d) If T were diagonalizable, T/U would be too (Ex. 60). So T is not diagonalizable.

§3.3 Duality (Ch. 10)

15. Dual bases.

a) If $\sum c_j\varphi_j = 0$, evaluate at v_k: $\sum c_j\varphi_j(v_k) = c_k = 0$.
b) n independent vectors in V', which has dimension $\dim \mathcal{L}(V, \mathbb{F}) = n$.
c) Both sides are linear functionals; at v_k the right side is $\sum_j \psi(v_j)\varphi_j(v_k) = \psi(v_k)$. Linear maps that agree on a basis are equal.
d) $B = \left(\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right)$, $B^{-1} = \left(\begin{smallmatrix} -2 & 3/2 \\ 1 & -1/2 \end{smallmatrix}\right)$: $\varphi_1(x, y) = -2x + \frac{3}{2}y$, $\varphi_2(x, y) = x - \frac{1}{2}y$. Check: $\varphi_1(1, 2) = 1$, $\varphi_1(3, 4) = 0$, $\varphi_2(1, 2) = 0$, $\varphi_2(3, 4) = 1$.

16. Functionals on $\mathcal{P}_2(\mathbb{R})$.

a) For $p = a + b(x-1) + c(x-1)^2$: $p(1) = a$, $p'(1) = b$, $\frac{1}{2}p''(1) = c$. So φ_i takes the ith coefficient, which is the defining property of the dual basis.
b) $\psi = \sum \psi(\text{basis vector})\varphi_j$: $\psi(1) = 1$, $\psi(x-1) = -\frac{1}{2}$, $\psi\big((x-1)^2\big) = \frac{1}{3}$. So $\psi = \varphi_1 - \frac{1}{2}\varphi_2 + \frac{1}{3}\varphi_3$.
c) $\varepsilon(1) = 1$, $\varepsilon(x-1) = -1$, $\varepsilon\big((x-1)^2\big) = 1$: $\varepsilon = \varphi_1 - \varphi_2 + \varphi_3$.
d) Coordinates $(1, 0, 0)$, $(1, -\frac{1}{2}, \frac{1}{3})$, $(1, -1, 1)$. Subtract the first from the others: $(0, -\frac{1}{2}, \frac{1}{3})$, $(0, -1, 1)$; subtract twice the first of these from the second: $(0, 0, \frac{1}{3})$. Rank 3, so they are independent: yes, a basis.

17. The double dual.

a) $j_V(v)(\varphi + \lambda\psi) = (\varphi + \lambda\psi)(v) = \varphi(v) + \lambda\psi(v)$.
b) $j_V(v + \lambda w)(\varphi) = \varphi(v + \lambda w) = \varphi(v) + \lambda\varphi(w)$ for every φ.
c) If $v \neq 0$, extend v to a basis and let φ be 1 at v and 0 at the other basis vectors. Then $j_V(v)(\varphi) = 1$, so $j_V(v) \neq 0$.
d) $\dim V'' = \dim V' = \dim V$, and an injective linear map between spaces of the same finite dimension is an isomorphism.

18. The dual of differentiation.

a) $T1 = 0$, $Tx = 1$, $Tx^2 = 2x$: $\mathcal{M}(T) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{smallmatrix}\right)$, $\mathcal{M}(T') = \mathcal{M}(T)^T = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$.
b) For $p = a_0 + a_1x + a_2x^2$, $p' = a_1 + 2a_2x$. $T'(\psi_0)(p) = \psi_0(p') = a_1$, so $T'(\psi_0) = \varphi_1$. $T'(\psi_1)(p) = 2a_2$, so $T'(\psi_1) = 2\varphi_2$.
c) T is surjective, so $\operatorname{null} T' = (\operatorname{range} T)^0 = \{0\}$. $\operatorname{range} T' = \operatorname{span}(\varphi_1, \varphi_2)$, the functionals that are 0 on the constants, which is $(\operatorname{null} T)^0$ since $\operatorname{null} T = \operatorname{span}(1)$.
d) $T'(\varepsilon)(p) = p'(1) = a_1 + 2a_2$: $T'(\varepsilon) = \varphi_1 + 2\varphi_2$.

19. Properties of the dual map.

a) $(S + T)'(\psi) = \psi \circ (S + T) = \psi \circ S + \psi \circ T$; $(cT)'(\psi) = \psi \circ (cT) = c(\psi \circ T)$.
b) $(ST)'(\psi) = \psi \circ S \circ T = (S'\psi) \circ T = T'(S'\psi)$.
c) By (b), $T'(T^{-1})' = (T^{-1}T)' = I_V' = I_{V'}$ and $(T^{-1})'T' = (TT^{-1})' = I_{W'}$.
d) $\mathcal{M}(T) = \left(\begin{smallmatrix} 1 & 1 \\ 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$, so $\mathcal{M}(T') = \left(\begin{smallmatrix} 1 & 2 & 0 \\ 1 & 0 & 3 \end{smallmatrix}\right)$.

20. Computing annihilators.

a) (a, b, c, d) is 0 on U iff $a + c = 0$ and $b + d = 0$. Basis: $(1, 0, -1, 0)$, $(0, 1, 0, -1)$.
b) $\varepsilon_1, \varepsilon_2 \in U^0$. They are independent: if $a\varepsilon_1 + b\varepsilon_2 = 0$, apply it to $x - 2$ to get $-a = 0$, and to $x - 1$ to get $b = 0$. U is the null space of $p \mapsto (p(1), p(2))$, which is surjective, so $\dim U = 2$ and $\dim U^0 = 4 - 2 = 2$. Hence $U^0 = \operatorname{span}(\varepsilon_1, \varepsilon_2)$.
c) No: $(\varepsilon_1 - \varepsilon_2)(x) = 1 - 2 = -1 \neq 0$.
d) $\dim W^0 = 4 - 2 = 2$. Example: $p \mapsto p''(0)$ (it is 0 on 1 and x, and 2 on x^2).

21. Null space and range of T'.

a) $T'\psi = 0$ iff $\psi(Tv) = 0$ for all v iff ψ is 0 on $\operatorname{range} T$.
b) If $v \in \operatorname{null} T$, $(T'\psi)(v) = \psi(Tv) = \psi(0) = 0$.
c) $\dim \operatorname{range} T' = \dim W - \dim \operatorname{null} T' = \dim W - (\dim W - \operatorname{rank} T) = \operatorname{rank} T = \dim V - \dim \operatorname{null} T = \dim(\operatorname{null} T)^0$. A subspace of the same dimension is the whole space.
d) T surjective iff $\operatorname{range} T = W$ iff $(\operatorname{range} T)^0 = \{0\}$ iff $\operatorname{null} T' = \{0\}$. From (c), $\operatorname{rank} T' = \operatorname{rank} T$; with $\mathcal{M}(T') = \mathcal{M}(T)^T$, $\operatorname{rank} A^T = \operatorname{rank} A$.

22. Annihilators in general.

a) Surjective: a functional on U extends to V (extend a basis of U, set it to 0 on the new vectors). Null space: $\varphi|_U = 0$ iff $\varphi \in U^0$. Rank–nullity: $\dim V' = \dim U^0 + \dim U'$, so $\dim U^0 = \dim V - \dim U$.

b) If $U \subseteq W$, anything 0 on W is 0 on U. Conversely, if $u \in U \setminus W$, then a basis of W together with u is independent; extend it to a basis of V and let φ be 1 at u and 0 on the other basis vectors. Then $\varphi \in W^0$ but $\varphi \notin U^0$.

c) φ is 0 on all $u + w$ iff it is 0 on U (take $w = 0$) and on W (take $u = 0$).

d) $v \mapsto \big(\varphi_1(v), \dots, \varphi_m(v)\big)$ is a linear map $V \to \mathbb{F}^m$ with null space $\bigcap \operatorname{null} \varphi_i = \{0\}$, so it is injective and $\dim V \le m$.

23. When functionals form a basis.

a) Let $\eta_1, \dots, \eta_n$ be the coordinate functionals on $\mathbb{F}^n$. $S'(\eta_i) = \eta_i \circ S = \varphi_i$, so $S'\big(\sum c_i \eta_i\big) = \sum c_i \varphi_i$. Hence S' is injective iff the φ_i are independent. And S' is injective iff S is surjective (Prop. 51).

b) n functionals in the n-dimensional V' form a basis iff they are independent iff S is surjective iff S is injective ($\dim V = n = \dim \mathbb{F}^n$) iff $\operatorname{null} S = \{0\}$.

c) A polynomial of degree at most 2 with roots $0, 1, 2$ is 0. By (b), yes.

§3.4 Invariant subspaces and eigenvalues (Ch. 11)

24. Invariant subspaces of a 3×3 matrix.

a) The matrix $\left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 5 \end{smallmatrix}\right)$ is lower-triangular, so in the reversed basis e_3, e_2, e_1 it is upper-triangular with the same diagonal (Prop. 60): eigenvalues $2, 5$. $E(2, T)$: $(T - 2I)(x, y, z) = (0, x, 3z) = 0$ iff $x = z = 0$: $\operatorname{span}(e_2)$. $E(5, T) = \operatorname{span}(e_3)$.

b) $Te_1 = (2, 1, 0) \notin \operatorname{span}(e_1)$: not invariant. $\operatorname{span}(e_1, e_2)$: $Te_1 = 2e_1 + e_2$, $Te_2 = 2e_2$: invariant. $\operatorname{span}(e_2, e_3)$: $Te_2 = 2e_2$, $Te_3 = 5e_3$: invariant.

c) An invariant line is spanned by an eigenvector: $\operatorname{span}(e_2)$ and $\operatorname{span}(e_3)$ only.

d) $\dim E(2, T) + \dim E(5, T) = 2 < 3$: not diagonalizable.

25. Existence of eigenvalues.

a) $v, Tv, \dots, T^n v$ are $n + 1$ vectors in an n-dimensional space, so $\sum_{k=0}^{n} a_k T^k v = 0$ with not all $a_k = 0$; take $p = \sum a_k x^k$. It is not a nonzero constant, since $a_0 v = 0$ with $v \neq 0$ forces $a_0 = 0$.

b) Over $\mathbb{C}$ the nonconstant p has a root λ (Fundamental Theorem of Algebra), so $p = (x - \lambda)q$ with $\deg q = \deg p - 1$. If $q(T)v = 0$, then q (nonzero, since $p \neq 0$) would be a smaller-degree polynomial with this property, contradicting minimality.

c) $(T - \lambda I)w = (T - \lambda I)q(T)v = p(T)v = 0$ and $w \neq 0$.

d) For R, $v = e_1$: $R^2 e_1 = -e_1$, so the minimal p is $x^2 + 1$, which has no real root; step (b) fails.

26. Independence of eigenvectors.

a) Suppose not; let k be smallest with $v_k = \sum_{i<k} a_i v_i$. Apply $T - \lambda_k I$: $0 = \sum_{i<k} a_i(\lambda_i - \lambda_k)v_i$. The v_i, $i < k$, are independent and $\lambda_i \neq \lambda_k$, so all $a_i = 0$, so $v_k = 0$, impossible for an eigenvector.

b) One eigenvector per eigenvalue gives an independent list, of length at most $\dim V$.

c) They are eigenvectors of differentiation (on infinitely differentiable functions) with distinct eigenvalues $1, 2, 3$.

d) $(1, \lambda, \lambda^2, \dots) \mapsto (\lambda, \lambda^2, \dots) = \lambda(1, \lambda, \dots)$. No contradiction: the space of sequences is infinite-dimensional.

27. A reflection of polynomials.

a) $T^2 p(x) = p(-(-x)) = p(x)$. If $Tp = \lambda p$, $p \neq 0$, then $p = T^2 p = \lambda^2 p$, so $\lambda^2 = 1$.

b) $Tp = p$ iff p is even: $E(1, T) = \operatorname{span}(1, x^2)$. $Tp = -p$ iff p is odd: $E(-1, T) = \operatorname{span}(x, x^3)$.

c) Dimensions $2 + 2 = 4 = \dim V$. Or: m_T divides $x^2 - 1 = (x - 1)(x + 1)$, which has distinct roots.

d) Same eigenvalues ± 1, each with eigenspace of dimension 2 (eigenvalues of the dual map).

28. Diagonalizing a 2×2 matrix.

a) $\chi_A(t) = t^2 - 5t + 6 = (t-2)(t-3)$. $A - 2I = \left(\begin{smallmatrix} 2 & -2 \\ 1 & -1 \end{smallmatrix}\right)$: eigenvector $(1,1)$. $A - 3I = \left(\begin{smallmatrix} 1 & -2 \\ 1 & -2 \end{smallmatrix}\right)$: eigenvector $(2,1)$.

b) $P = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 1 \end{smallmatrix}\right)$, $D = \operatorname{diag}(2,3)$, $P^{-1} = \left(\begin{smallmatrix} -1 & 2 \\ 1 & -1 \end{smallmatrix}\right)$.

c) $A^k = PD^kP^{-1} = \left(\begin{smallmatrix} -2^k + 2\cdot 3^k & 2^{k+1} - 2\cdot 3^k \\ -2^k + 3^k & 2^{k+1} - 3^k \end{smallmatrix}\right)$. Check $k = 1$: A.

d) Distinct eigenvalues $2, 3$, diagonalizable: $m_A = (x-2)(x-3)$.

29. Eigenvalues that survive changes.

a) $Rv = \lambda v \iff S^{-1}TSv = \lambda v \iff T(Sv) = \lambda(Sv)$, and $Sv \neq 0 \iff v \neq 0$.

b) $(T - \lambda I)' = T' - \lambda I$, and a map and its dual have the same rank; V and V' have the same dimension, so the null spaces have the same dimension.

c) $E(1, A) = \operatorname{span}(e_1)$ and $E(1, A^T) = \operatorname{span}(e_2)$: same dimension, different subspaces.

30. Stochastic matrices.

a) With $\|x\|_1 = \sum |x_i|$: $\|Px\|_1 \leq \sum_i \sum_j P_{ij}|x_j| = \sum_j |x_j|$ (columns sum to 1). If $Px = \lambda x$, $x \neq 0$: $|\lambda|\|x\|_1 \leq \|x\|_1$.

b) Column sums 1 mean $\mathbf{1}^T P = \mathbf{1}^T$, i.e. $P^T\mathbf{1} = \mathbf{1}$. So 1 is an eigenvalue of P^T, and P and P^T have the same eigenvalues (the matrix of the dual map is the transpose).

c) $(P - I)\pi = 0$: $-\frac{1}{2}\pi_1 + \frac{1}{5}\pi_2 = 0$, so $\pi_2 = \frac{5}{2}\pi_1$. Normalizing: $\pi = (\frac{2}{7}, \frac{5}{7})$.

d) $\chi_P(t) = t^2 - \frac{13}{10}t + \frac{3}{10} = (t-1)(t - \frac{3}{10})$: the other eigenvalue is $\frac{3}{10}$.

31. PageRank.

a) If $(I - \alpha P)x = 0$ with $x \neq 0$, then $Px = \frac{1}{\alpha}x$ with $\frac{1}{\alpha} > 1$, contradicting Q30(a). So $I - \alpha P$ is injective, hence invertible.

b) $r_1 = \frac{1}{2}r_2 + \frac{1}{2}$ and $r_2 = \frac{1}{2}r_1$. So $r_1 = \frac{1}{4}r_1 + \frac{1}{2}$, $r_1 = \frac{2}{3}$, $r_2 = \frac{1}{3}$.

c) Entries are nonnegative and add to 1. All teleports go to page 1, and page 1 also receives links from page 2, so page 1 ranks higher.

32. Eigenvalues of restriction and quotient.

a) An eigenvector of $T|_U$ is a nonzero $u \in U$ with $Tu = \lambda u$.

b) $v \notin U$ means $v + U \neq 0 + U$, and $(T/U)(v + U) = Tv + U = \lambda(v + U)$.

c) $T - \lambda I$ is injective, so its restriction to U is injective; it maps U into U (invariance), so onto U (same finite dimension). If $\big((T/U) - \lambda I\big)(v + U) = 0 + U$, then $(T - \lambda I)v \in U$, so $(T - \lambda I)v = (T - \lambda I)u$ for some $u \in U$, and injectivity gives $v = u \in U$, i.e. $v + U = 0 + U$. So λ is not an eigenvalue of T/U.

§3.5 Upper-triangular matrices and diagonalizability (Ch. 12)

33. Upper-triangular matrices.

a) Column j of $\mathcal{M}(T)$ holds the coefficients of Tv_j; zeros below row j mean $Tv_j = \sum_{i \leq j} a_{ij}v_i$.

b) For $i \leq j$, $Tv_i \in \operatorname{span}(v_1, \ldots, v_i) \subseteq \operatorname{span}(v_1, \ldots, v_j)$, so T maps a spanning list of $\operatorname{span}(v_1, \ldots, v_j)$ into it.

c) $v_j \in \operatorname{span}(v_1, \ldots, v_j)$, which is invariant, so $Tv_j \in \operatorname{span}(v_1, \ldots, v_j)$: the entries of column j below row j are 0.

34. Eigenvalues on the diagonal.

a) $\mathcal{M}(T - \lambda I)$ is upper-triangular with diagonal $\lambda_k - \lambda$. If $\lambda \neq$ every λ_k, back-substitution in $(T - \lambda I)x = 0$ (last row first) gives $x = 0$: injective, not an eigenvalue. If $\lambda = \lambda_k$, the first k columns of $\mathcal{M}(T - \lambda I)$ have zeros in rows $k, \ldots, n$, so they are k vectors in a $(k-1)$-dimensional space, hence dependent: not injective.

b) T invertible iff 0 is not an eigenvalue iff no $\lambda_k = 0$.

c) In $1, x, x^2$: $\mathcal{M}(T) = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, so the only eigenvalue is 1. $E(1, T) = \operatorname{null} D = \operatorname{span}(1)$.

d) $D^3 = 0$ on $\mathcal{P}_2(\mathbb{R})$, and $(I + D)(I - D + D^2) = I + D^3 = I$. So $T^{-1} = I - D + D^2$.

35. Triangularizing a matrix.

a) $\chi_A(t) = t^2 - 2t + 1 = (t-1)^2$: eigenvalue 1. $A - I = \left(\begin{smallmatrix} 2 & 1 \\ -4 & -2 \end{smallmatrix}\right)$: eigenvectors are the multiples of $(1, -2)$.

b) $s_1 = (1, -2)$, $s_2 = e_2$. $As_1 = s_1$ and $As_2 = (1, -1) = s_1 + s_2$. So with $S = \left(\begin{smallmatrix} 1 & 0 \\ -2 & 1 \end{smallmatrix}\right)$, $S^{-1}AS = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$.

c) $\dim E(1, A) = 1 < 2$, so not diagonalizable. $(A - I)^2 = 0$ (multiply out) and $A \neq I$, so $m_A = (x-1)^2$.

36. A diagonalizability test with a parameter.

a) Lower-triangular with diagonal $1, 2, 2$; in the reversed basis e_3, e_2, e_1 it is upper-triangular (Prop. 60): eigenvalues 1 and 2.

b) $E(1)$ has dimension 1 ($A_t - I$ has rank 2). $A_t - 2I = \left(\begin{smallmatrix} -1 & 0 & 0 \\ 1 & 0 & 0 \\ -3 & t & 0 \end{smallmatrix}\right)$ has rank 1 if $t = 0$ and rank 2 if $t \neq 0$. So $\dim E(1) + \dim E(2) = 3$ iff $t = 0$: diagonalizable exactly for $t = 0$.

c) $t = 0$: $E(1)$ is the null space of $\left(\begin{smallmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ -3 & 0 & 1 \end{smallmatrix}\right)$, spanned by $(1, -1, 3)$. $E(2) = \{x = 0\} = \operatorname{span}(e_2, e_3)$. Eigenbasis $(1, -1, 3),\ e_2,\ e_3$.

37. Conditions for diagonalizability.

a) Column j of a diagonal matrix has only the (j, j) entry, i.e. $Tv_j = a_{jj}v_j$: each basis vector is an eigenvector, and conversely.

b) If $w_1 + \cdots + w_m = 0$ with $w_i \in E(\lambda_i, T)$, the nonzero w_i would be eigenvectors for distinct eigenvalues with a nontrivial dependence, contradicting independence (Prop. 54). So all $w_i = 0$.

c) ($\Rightarrow$) The eigenbasis lies in the sum, so the (direct) sum is V. ($\Leftarrow$) Put together bases of the $E(\lambda_i, T)$.

d) The direct sum has dimension $\sum \dim E(\lambda_i, T)$, and a subspace equals V iff its dimension is $\dim V$.

38. Projections.

a) $v = (v - Pv) + Pv$ with $P(v - Pv) = Pv - P^2v = 0$. If $v = Pw \in \operatorname{range} P$ and $Pv = 0$, then $v = Pw = P^2w = Pv = 0$.

b) If $v = Pw$ then $Pv = P^2w = Pw = v$; if $Pv = v$ then $v \in \operatorname{range} P$.

c) $V = E(0, P) \oplus E(1, P)$ by (a),(b), so P is diagonalizable. m_P divides $x(x - 1)$: it is x ($P = 0$), $x - 1$ ($P = I$), or $x(x - 1)$ (otherwise).

d) $(x, y) = y(1, 1) + (x - y)(1, 0)$, so $P(x, y) = (y, y)$: matrix $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)$.

39. A recurrence.

a) $A = \left(\begin{smallmatrix} 0 & 1 \\ 2 & 1 \end{smallmatrix}\right)$, $\chi_A(t) = t^2 - t - 2 = (t - 2)(t + 1)$. Eigenvectors: $(1, 2)$ for 2, $(1, -1)$ for -1.

b) $(0, 1) = \frac{1}{3}(1, 2) - \frac{1}{3}(1, -1)$.

c) $A^n(0, 1) = \frac{1}{3}\big(2^n(1, 2) - (-1)^n(1, -1)\big)$; the first entry is $a_n = \dfrac{2^n - (-1)^n}{3}$. Check: $a_2 = \frac{4-1}{3} = 1$, $a_3 = \frac{8+1}{3} = 3$; and indeed $a_2 = 1 + 0 = 1$, $a_3 = 1 + 2 = 3$.

40. A two-state Markov chain.

a) $\chi_P(t) = t^2 - 1.7t + 0.7 = (t - 1)(t - 0.7)$. Eigenvectors: $(2, 1)$ for 1 ($-0.1x + 0.2y = 0$), $(1, -1)$ for 0.7.

b) $\pi = (\frac{2}{3}, \frac{1}{3})$, and $(1, 0) = \pi + \frac{1}{3}(1, -1)$.

c) $P^k p_0 = \pi + \frac{1}{3}(0.7)^k(1, -1)$.

d) The limit is $\pi = (\frac{2}{3}, \frac{1}{3})$; the error is multiplied by 0.7 at each step.

§3.6 The minimal polynomial (Ch. 13)

41. Minimal polynomials from the definition.

a) A is not a multiple of I, so I, A are independent. $A^2 = \left(\begin{smallmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 2A - I$. So $m_A = x^2 - 2x + 1 = (x - 1)^2$.

b) B is not a multiple of I; $(B - I)(B - 2I) = \operatorname{diag}(0, 0, 0)$, so $m_B = (x - 1)(x - 2)$.

c) B (distinct roots); not A (repeated root).

d) $m_{A-I}(x) = m_A(x + 1) = x^2$. m_{2B}: $2B = \operatorname{diag}(2, 2, 4)$, so $m_{2B} = (x - 2)(x - 4)$.

42. Polynomials that annihilate T.

a) If $q = m_T r$, then $q(T) = m_T(T)r(T) = 0$. Conversely, divide: $q = m_T r + s$ with $s = 0$ or $\deg s < \deg m_T$. Then $s(T) = q(T) - m_T(T)r(T) = 0$, and minimality forces $s = 0$.

b) $x^4 - x = x(x - 1)(x^2 + x + 1)$ has the distinct roots $0, 1, \omega, \bar{\omega}$ ($\omega = e^{2\pi i/3}$). m_T divides it, so m_T has distinct roots and T is diagonalizable; eigenvalues lie among $0, 1, \omega, \bar{\omega}$.

c) Rotation by 120°: $R^3 = I$, so $R^4 = R$, but R has no real eigenvalue, so it is not diagonalizable over $\mathbb{R}$. Real eigenvalues must be real roots of $x^4 - x$: only 0 and 1.

43. The degree bound.

a) $v, \dots, T^n v$ are dependent, giving nonzero q of degree $\le n$ with $q(T)v = 0$. If $q(T) = 0$, then m_T divides q and $\deg m_T \le n$.

b) If $q(T)u = 0$ then $q(T)Tu = Tq(T)u = 0$. $v \in U$, so $U \neq \{0\}$; $q(T) \neq 0$, so $U \neq V$.

c) $m_{T/U}(T)v \in U$ for every v (since $m_{T/U}(T/U) = 0$ and $p(T/U)(v+U) = p(T)v + U$), and $m_{T|_U}(T)$ sends U to 0. So $m_{T|_U}(T)m_{T/U}(T) = 0$, and Q42(a) applies.

d) Base case $n = 1$: $V = \operatorname{span}(v)$, so $q(T)v = 0$ gives $q(T) = 0$ and (a) applies. Inductive step: by induction $\deg m_{T|_U} \le \dim U$ and $\deg m_{T/U} \le \dim V/U$. So $\deg m_T \le \dim U + \dim V/U = n$.

44. Distinct roots give diagonalizability.

a) Divide g by $x - \lambda_1$: the remainder s has degree < 1, so it is a constant, and $s = g(\lambda_1) = \prod_{i \ge 2}(\lambda_1 - \lambda_i) \neq 0$.

b) $(T - \lambda_1 I)u = \frac{1}{s} m_T(T)v = 0$. And $v - u = \frac{1}{s}\big(sv - g(T)v\big) = -\frac{1}{s}(T - \lambda_1 I)q(T)v \in \operatorname{range}(T - \lambda_1 I)$.

c) By (b) V is the sum. By rank–nullity the dimensions add to $\dim V$, so the sum is direct.

d) $U = \operatorname{range}(T - \lambda_1 I)$ is invariant, $m_{T|_U}$ divides m_T (so has distinct roots), and $\dim U < \dim V$ since λ_1 is an eigenvalue. By induction $T|_U$ has an eigenbasis; add a basis of $\operatorname{null}(T - \lambda_1 I)$. (Base case: $m = 1$ means $T = \lambda_1 I$.)

45. A companion matrix.

a) $Ae_1 = (0,0,6)$ (the first column), $A^2e_1 = A(0,0,6) = (0,6,36)$, $A^3e_1 = A(0,6,36) = (6,36,150)$. The vectors $e_1, (0,0,6), (0,6,36)$ are independent (look at the positions of the first nonzero entries), so they form a basis. If $q(A) = 0$ with $\deg q \le 2$, then $q(A)e_1 = 0$ is a dependence among e_1, Ae_1, A^2e_1, so $q = 0$; hence $\deg m_A \ge 3$. Solving $A^3e_1 = ae_1 + bAe_1 + cA^2e_1 = (a, 6c, 6b + 36c)$ gives $a = 6$, $c = 6$, $b = -11$. So $p(x) = x^3 - 6x^2 + 11x - 6$ has $p(A)e_1 = 0$, hence $p(A)A^je_1 = A^jp(A)e_1 = 0$ for each basis vector, so $p(A) = 0$ and $m_A = p = (x-1)(x-2)(x-3)$.

b) Yes: m_A splits over $\mathbb{R}$ with distinct roots.

c) $A(A^2 - 6A + 11I) = 6I$, so $A^{-1} = \frac{1}{6}(A^2 - 6A + 11I)$.

d) $x^4 = (x+6)m_A(x) + 25x^2 - 60x + 36$, so $A^4 = 25A^2 - 60A + 36I$.

46. Minimal polynomial of a direct sum.

a) Write $v = u + w$. Then $p(T)v = p(T|_U)u + p(T|_W)w$ with the two terms in U and W. So $p(T) = 0$ iff both restrictions give 0 (the sum is direct).

b) By (a) and Q42(a), $p(T) = 0$ iff $m_{T|_U}$ and $m_{T|_W}$ both divide p. m_T is the monic polynomial of least degree with $m_T(T) = 0$, so it is the monic polynomial of least degree divisible by both.

c) If both restrictions have minimal polynomials with distinct linear factors, so does their least common multiple, and conversely each divides m_T.

d) The blocks have minimal polynomials $(x-1)^2$ and $(x-1)(x-2)$. Least common multiple: $m_T = (x-1)^2(x-2)$.

47. An operator on polynomials.

a) $T(x^k) = kx^k$: $\mathcal{M}(T) = \operatorname{diag}(0,1,2,3)$. Eigenvalues $0, 1, 2, 3$.

b) Diagonalizable with distinct eigenvalues: $m_T = x(x-1)(x-2)(x-3)$.

c) $T^2 = \operatorname{diag}(0,1,4,9)$: $m_{T^2} = x(x-1)(x-4)(x-9)$.

d) The Lagrange polynomial that is 1 at 2 and 0 at $0, 1, 3$: $p(x) = \dfrac{x(x-1)(x-3)}{2 \cdot 1 \cdot (-1)} = -\frac{1}{2}x(x-1)(x-3)$. Then $p(T)x^k = p(k)x^k$, which is x^2 for $k = 2$ and 0 otherwise.

48. Real operators.

a) Factor m_T over $\mathbb{R}$. If it has a factor $x - \lambda$, then λ is an eigenvalue and an eigenvector spans an invariant line. Otherwise $m_T = (x^2 + bx + c)q$; minimality gives v with $w = q(T)v \neq 0$, and $(T^2 + bT + cI)w = 0$, so $\operatorname{span}(w, Tw)$ is invariant, of dimension 1 or 2.

b) Dimension 1: clear. If $\dim V \ge 3$ is odd and T has no eigenvalue, (a) gives an invariant U of dimension 2. Then $\dim V/U$ is odd and smaller, so T/U has an eigenvalue λ, a root of $m_{T/U}$, which divides m_T; so λ is a root of m_T and an eigenvalue of T. Contradiction.

c) $T^2 = -I$, so an eigenvalue would satisfy $\lambda^2 = -1$: none real. $\operatorname{span}(e_1, e_2)$ is invariant ($Te_1 = e_2$, $Te_2 = -e_1$).

§4 Practice midterms

§4.1 Practice Midterm A

1.

a) (a,b,c) is 0 on U iff $a+2b=0$ and $b+c=0$, so $(a,b,c)=c(2,-1,1)$. Basis of U^0: $(2,-1,1)$. (Check: $\dim U^0 = 3-2=1$.)

b) Put v_1,v_2,v_3 as columns of $B=\left(\begin{smallmatrix}1&0&1\\1&1&0\\0&1&1\end{smallmatrix}\right)$. The dual basis is the rows of $B^{-1}=\frac{1}{2}\left(\begin{smallmatrix}1&1&-1\\-1&1&1\\1&-1&1\end{smallmatrix}\right)$: $\varphi_1=\frac{1}{2}(x+y-z)$, $\varphi_2=\frac{1}{2}(-x+y+z)$, $\varphi_3=\frac{1}{2}(x-y+z)$. Check one: $\varphi_1(v_1)=1$, $\varphi_1(v_2)=0$, $\varphi_1(v_3)=0$.

c) $\psi=\sum\psi(v_j)\varphi_j$ with $\psi(v_1)=-1$, $\psi(v_2)=1$, $\psi(v_3)=4$: $\psi=-\varphi_1+\varphi_2+4\varphi_3$. Check: $\frac{1}{2}\big(-(x+y-z)+(-x+y+z)+4(x-y+z)\big)=x-2y+3z$.

2.

a) $Tv=\lambda v$, $v\neq 0$: $0=(T^3-T)v=(\lambda^3-\lambda)v$, so $\lambda(\lambda-1)(\lambda+1)=0$.

b) $a+b+c=v-T^2v+T^2v=v$. $Ta=Tv-T^3v=0$. $Tb=\frac{1}{2}(T^3v+T^2v)=\frac{1}{2}(Tv+T^2v)=b$. $Tc=\frac{1}{2}(T^3v-T^2v)=\frac{1}{2}(Tv-T^2v)=-c$.

c) By (b), $V=\operatorname{null}T+E(1,T)+E(-1,T)$. Eigenspaces for distinct eigenvalues form a direct sum, so V is their direct sum and T is diagonalizable (LADR 5.55).

d) $T^2(x,y,z)=T(0,z,y)=(0,y,z)$ and $T^3(x,y,z)=(0,z,y)=T(x,y,z)$. $\operatorname{null}T=\operatorname{span}(e_1)$; $E(1,T)=\operatorname{span}((0,1,1))$; $E(-1,T)=\operatorname{span}((0,1,-1))$. All three eigenvalues occur and T is diagonalizable, so $m_T=x(x-1)(x+1)$.

3.

a) Block form: the top-left block $\left(\begin{smallmatrix}1&1\\-1&3\end{smallmatrix}\right)$ has $\chi=t^2-4t+4=(t-2)^2$, and the last entry is 2. Only eigenvalue 2. $A-2I=\left(\begin{smallmatrix}-1&1&0\\-1&1&0\\0&0&0\end{smallmatrix}\right)$ has rank 1, so $\dim E(2,A)=2$.

b) $(A-2I)^2=0$ (direct multiplication; the top-left block $\left(\begin{smallmatrix}-1&1\\-1&1\end{smallmatrix}\right)$ squares to 0), and $A-2I\neq 0$, so $m_A=(x-2)^2$. Not diagonalizable (repeated root, or $2<3$ in the dimension test).

c) $A=2I+N$ with $N=A-2I$, $N^2=0$, and $2I$ commuting with N. So $A^k=2^kI+k2^{k-1}N$:

$$A^k=\begin{pmatrix}2^k-k2^{k-1}&k2^{k-1}&0\\-k2^{k-1}&2^k+k2^{k-1}&0\\0&0&2^k\end{pmatrix}.$$

4.

a) If $v-v'\in U$, then $Tv-Tv'=T(v-v')\in U$ by invariance, so $Tv+U=Tv'+U$. Linearity is inherited from T.

b) $m_T(T|_U)=m_T(T)|_U=0$, and $m_T(T/U)(v+U)=m_T(T)v+U=0+U$. So both minimal polynomials divide m_T (Prop. 65).

c) Write $r=m_{T|_U}$, $s=m_{T/U}$. $s(T/U)=0$ means $s(T)v\in U$ for every v. Then $r(T)s(T)v=r(T|_U)\big(s(T)v\big)=0$. So $(rs)(T)=0$ and m_T divides rs.

d) If T is diagonalizable, m_T is a product of distinct linear factors, and so is its divisor $m_{T|_U}$; hence $T|_U$ is diagonalizable. Example: $T=\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$, $U=\operatorname{span}(e_1)$: $T|_U=0$ and $T/U=0$, but T is not diagonalizable.

§4.2 Practice Midterm B

1.

a) $T(1)=(1,0)$, $T(x)=(0,1)$, $T(x^2)=(0,2)$. $\mathcal{M}(T)=\left(\begin{smallmatrix}1&0&0\\0&1&2\end{smallmatrix}\right)$.

b) $\mathcal{M}(T')=\mathcal{M}(T)^T=\left(\begin{smallmatrix}1&0\\0&1\\0&2\end{smallmatrix}\right)$, so $T'(\psi_1)=\varphi_0$ and $T'(\psi_2)=\varphi_1+2\varphi_2$. Directly: $T'(\psi_2)(p)=p'(1)=a_1+2a_2$ for $p=a_0+a_1x+a_2x^2$.

c) T is surjective ($T(1)=e_1$, $T(x)=e_2$), so T' is injective: $\operatorname{null}T'=\{0\}$. $\operatorname{null}T$: $a_0=0$ and $a_1+2a_2=0$, basis x^2-2x. $\dim(\operatorname{null}T)^0=3-1=2=\operatorname{rank}T'$. Consistent.

2.

a) $Tv=\lambda v\Rightarrow T(Sv)=STv=\lambda Sv$.

b) T has an eigenvalue λ (complex, $V\neq\{0\}$). $E(\lambda,T)\neq\{0\}$ is S-invariant, so $S|_{E(\lambda,T)}$ has an eigenvector w. Then $Sw=\mu w$ and $Tw=\lambda w$.

c) $V = \bigoplus_i E(\lambda_i, T)$. Each $E(\lambda_i, T)$ is S-invariant, and S restricted to it is diagonalizable. Choose an eigenbasis of $S|_{E(\lambda_i,T)}$ in each; every vector in it is an eigenvector of both. Together they form a basis of V.

d) For $j = 1, 2, 3$, $E(j, T) = \operatorname{span}(e_j)$ is 1-dimensional and S-invariant, so $Se_j \in \operatorname{span}(e_j)$ for each j: S is diagonal.

3.

a) $T(A + cB) = (A + cB)x_0 = Ax_0 + cBx_0$ with $x_0 = (1, -1)$, so T is linear. $T(A) = (a_{11} - a_{12},\ a_{21} - a_{22})$; $T(E_{11}) = e_1$ and $T(E_{21}) = e_2$, so T is surjective. $N = \{a_{11} = a_{12},\ a_{21} = a_{22}\}$, with basis $\left(\begin{smallmatrix}1&1\\0&0\end{smallmatrix}\right), \left(\begin{smallmatrix}0&0\\1&1\end{smallmatrix}\right)$ (dimension $4 - 2 = 2$).

b) $A + N \mapsto A\left(\begin{smallmatrix}1\\-1\end{smallmatrix}\right)$. Basis of V/N: $E_{11} + N,\ E_{21} + N$, which map to e_1, e_2.

c) If $A \in N$, then $(BA)x_0 = B(Ax_0) = B0 = 0$, so $BA \in N$. Write $\Phi\colon V/N \to \mathbb{R}^2$ for the isomorphism of (b). Then $\Phi\big((L/N)(A + N)\big) = \Phi(BA + N) = BAx_0 = B\,\Phi(A + N)$. So $\Phi \circ (L/N) \circ \Phi^{-1}$ is $x \mapsto Bx$, and L/N has the same eigenvalues as B: 2 and 3 (triangular).

4.

a) If $Tv = \lambda v$ with $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$. Conversely, if $m_T(\lambda) = 0$, write $m_T = (x - \lambda)q$ with $\deg q < \deg m_T$, so $q(T) \neq 0$ by minimality: pick v with $w = q(T)v \neq 0$. Then $(T - \lambda I)w = m_T(T)v = 0$, so w is an eigenvector for λ.

b) Let $q(x) = m_T(x + \lambda)$; it is monic of degree $\deg m_T$, and $q(T - \lambda I) = m_T\big((T - \lambda I) + \lambda I\big) = m_T(T) = 0$. If some nonzero p of smaller degree had $p(T - \lambda I) = 0$, then $r(x) = p(x - \lambda)$ would have $r(T) = p(T - \lambda I) = 0$ and $\deg r = \deg p < \deg m_T$, which is impossible. So $m_{T-\lambda I} = q$.

c) $(x - 1)(x - 2)$ has distinct roots and sends T to 0, so T is diagonalizable (T2), and m_T divides $(x - 1)(x - 2)$. m_T is not $x - 1$ or $x - 2$ (since $T \neq I, 2I$), so $m_T = (x - 1)(x - 2)$, and both 1 and 2 are eigenvalues (Prop. 66). With $a = \dim E(1, T) \geq 1$, $b = \dim E(2, T) \geq 1$ and $a + b = 4$: $(a, b) \in \{(1, 3), (2, 2), (3, 1)\}$, and each occurs (diag matrices).

§4.3 Practice Midterm C

1.

a) $T(1) = 0$, $T(x) = x + 1$, $T(x^2) = 2x(x + 1) = 2x^2 + 2x$. $\mathcal{M}(T) = \left(\begin{smallmatrix}0&1&0\\0&1&2\\0&0&2\end{smallmatrix}\right)$, upper-triangular: eigenvalues $0, 1, 2$.

b) Three distinct eigenvalues on a 3-dimensional space, so diagonalizable. Eigenvectors: $T(1) = 0$; $T(x+1) = x+1$; $T\big((x + 1)^2\big) = (x + 1) \cdot 2(x + 1) = 2(x + 1)^2$. Eigenbasis $1,\ x + 1,\ (x + 1)^2$.

c) $x^2 = (x + 1)^2 - 2(x + 1) + 1$. For $k \geq 1$: $T^k(x^2) = 2^k(x + 1)^2 - 2(x + 1)$ (the eigenvalue-0 part dies).

2.

a) With bases $v_1, \ldots, v_k$ of V and $w_1, \ldots, w_l$ of W, the list $(v_1, 0), \ldots, (v_k, 0), (0, w_1), \ldots, (0, w_l)$ spans ($(v, w) = (v, 0) + (0, w)$) and is independent ($\big(\sum a_i v_i, \sum b_j w_j\big) = (0, 0)$ forces all coefficients 0). So $\dim(V \times W) = k + l$.

b) $\Gamma(u_1, u_2) = 0$ iff $u_2 = -u_1$, which forces $u_1 \in U_1 \cap U_2$. So $\operatorname{null}\Gamma = \{(u, -u) : u \in U_1 \cap U_2\}$, and $u \mapsto (u, -u)$ is an isomorphism $U_1 \cap U_2 \to \operatorname{null}\Gamma$. Γ maps onto $U_1 + U_2$, so rank–nullity gives $\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$.

c) $U_1 = \operatorname{span}(e_1)$, $U_2 = \operatorname{span}(e_2)$, $U_3 = \operatorname{span}(e_1 + e_2)$: distinct lines, so they meet pairwise in $\{0\}$. $\big(e_1,\ e_2,\ -(e_1 + e_2)\big)$ is a nonzero vector in $\operatorname{null}\Gamma$, so Γ is not injective and the sum is not direct ($\dim(U_1 + U_2 + U_3) = 2 < 3$).

3.

a) $Ae_1 = e_2$, $A^2e_1 = e_3$, $A^3e_1 = Ae_3 = (2, -1, 2) = 2e_1 - e_2 + 2e_3$. So $(A^3 - 2A^2 + A - 2I)e_1 = 0$. e_1 is a cyclic vector (e_1, Ae_1, A^2e_1 is a basis), so $\deg m_A = 3$ and $p(A)$ sends each basis vector e_1, Ae_1, A^2e_1 to 0, where $p(x) = x^3 - 2x^2 + x - 2$ (since $p(A)A^je_1 = A^jp(A)e_1 = 0$). So $p(A) = 0$. Hence $m_A = x^3 - 2x^2 + x - 2 = (x - 2)(x^2 + 1)$.

b) Over $\mathbb{R}$: only 2 (the roots of $x^2 + 1$ are not real). Over $\mathbb{C}$: $2, i, -i$. Not diagonalizable over $\mathbb{R}$ (m_A does not split); diagonalizable over $\mathbb{C}$ (distinct roots).

c) Constant term $-2 \neq 0$, so A is invertible. $A^3 - 2A^2 + A = 2I$, i.e. $A(A^2 - 2A + I) = 2I$, so $A^{-1} = \frac{1}{2}(A^2 - 2A + I)$.

4.

a) $R\colon V' \to U'$, $\varphi \mapsto \varphi|_U$, is linear and surjective, with $\operatorname{null} R = U^0$. Rank–nullity: $\dim V' = \dim U^0 + \dim U'$, i.e. $\dim U^0 = \dim V - \dim U$.

b) φ is 0 on $U + W$ iff it is 0 on every $u + w$ iff it is 0 on U and on W (take $w = 0$ or $u = 0$; conversely add).

c) $\dim\operatorname{null} T' = \dim(\operatorname{range} T)^0 = \dim X - \dim\operatorname{range} T = \dim X - (\dim V - \dim\operatorname{null} T)$, using Prop. 50, the annihilator dimension formula, and rank–nullity.

d) φ is 0 on its own null space, so $\operatorname{span}(\varphi) \subseteq U^0$. Since $\varphi \neq 0$, $\operatorname{range}\varphi = \mathbb{F}$, so $\dim U = \dim V - 1$ and $\dim U^0 = 1$. A 1-dimensional space containing the line $\operatorname{span}(\varphi)$ equals it. For $U = \{p : p(1) = 0\} = \operatorname{null}\varphi$ with $\varphi(p) = p(1)$: basis of U^0 is φ, "evaluate at 1".

§5 Problem bank

Problem 1

(a) If $a \neq b$, the polynomial $(x-a)(x-b)$ has distinct roots and $(T-aI)(T-bI) = 0$, so T is diagonalizable (T2). If $a = b$ and T were diagonalizable, its only eigenvalue would be a, so $T = aI$ in an eigenbasis, which is excluded. So T is diagonalizable iff $a \neq b$.

(b) By (a), T is diagonalizable, and its only possible eigenvalues are a, b (T1). So $\mathbb{C}^n = E(a,T) \oplus E(b,T)$ (Prop. 62). Both a and b are eigenvalues (T1(b)), so both eigenspaces are nonzero, and $1 \leq \dim E(a,T) = n - \dim E(b,T) \leq n-1$.

(c) Not necessarily: $T = aI + N$ on $\mathbb{F}^2$, with $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, has $(T-aI)^2 = N^2 = 0$ but is not diagonalizable. In general T is diagonalizable iff $T = aI$ (then $m_T = x - a$); otherwise $m_T = (x-a)^2$ has a repeated root.

Problem 2

(a) $(T-\lambda I)'\varphi = \varphi \circ (T - \lambda I) = \varphi \circ T - \lambda\varphi = (T' - \lambda I)\varphi$.

(b) $E(\lambda, T') = \operatorname{null}(T' - \lambda I) = \operatorname{null}(T - \lambda I)' = (\operatorname{range}(T - \lambda I))^0$ by (a) and Prop. 50.

(c) $T'\varphi_1 = \varphi_1 \circ T = (x, y) \mapsto x + y$, so $T'\varphi_1 = \varphi_1 + \varphi_2$; $T'\varphi_2 = \varphi_2$. So $(T' - I)\varphi_1 = \varphi_2$, $(T' - I)\varphi_2 = 0$, and $E(1, T') = \operatorname{span}(\varphi_2)$. Check: $(T - I)(x, y) = (y, 0)$, so $\operatorname{range}(T - I) = \operatorname{span}(e_1)$, whose annihilator is $\operatorname{span}(\varphi_2)$.

(d) $\dim E(\lambda, T) = \dim E(\lambda, T')$ for every λ (Section 1, eigenvalues of the dual map), and $\dim V = \dim V'$. So the dimension test for diagonalizability gives the same answer for both.

Problem 3

(a) $Av = (1, 0, 2)$, $A^2v = (1, 0, 4) = 3Av - 2v$, and v, Av are independent. By T6 the smallest invariant subspace is $W = \operatorname{span}(v, Av) = \operatorname{span}(e_1, e_3)$, and the restriction has minimal polynomial $x^2 - 3x + 2 = (x-1)(x-2)$. For $u = (1, 1, 1)$: $Au = (2, 1, 2)$, $A^2u = (3, 1, 4)$, and the matrix with columns u, Au, A^2u row-reduces to rank 3, so u is cyclic and $\deg m_A = 3$ (T6). A is upper-triangular with diagonal $1, 1, 2$, so m_A has roots exactly $1, 2$, and it is $(x-1)^2(x-2)$ or $(x-1)(x-2)^2$. Compute $(A - I)^2 = \operatorname{diag}(0, 0, 1)$ and $(A - I)^2(A - 2I) = 0$ (the last row of $A - 2I$ is 0). So $m_A = (x-1)^2(x-2)$.

(b) $v, Tv, \ldots, T^{n-1}v$ is a basis, so $Sv = p(T)v$ for some polynomial p. For each j, $S(T^jv) = T^jSv = T^jp(T)v = p(T)(T^jv)$. So S and $p(T)$ agree on a basis: $S = p(T)$.

(c) I has no cyclic vector ($\operatorname{span}(v, Iv)$ is a line), and $S = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ commutes with I but is not a polynomial in I (those are the multiples cI).

Problem 4

(a) $T(1) = 0$, $T(x) = 0$, $T(x^2) = 2x^2 + 2$, $T(x^3) = 6x^3 + 6x$. $\mathcal{M}(T) = \left(\begin{smallmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \end{smallmatrix}\right)$, upper-triangular with diagonal $0, 0, 2, 6$. Eigenvalues $0, 2, 6$.

(b) $\operatorname{null} T = \operatorname{span}(1, x)$ has dimension 2; $E(2)$ and $E(6)$ have dimension ≥ 1. $2 + 1 + 1 = 4$, so diagonalizable. Eigenbasis: 1, x, $x^2 + 1$, $x^3 + x$ (check: $T(x^2 + 1) = 2x^2 + 2$; $T(x^3 + x) = 6x^3 + 6x$).

(c) $m_T = x(x-2)(x-6)$.

Problem 5

(a) $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathbb{R}^2$, $T(p) = \big(p(1), p(-1)\big)$, is linear and surjective ($T(1) = (1, 1)$, $T(x) = (1, -1)$, independent). So $\dim U = \dim\operatorname{null} T = 4 - 2 = 2$. Basis of U: $x^2 - 1$, $x^3 - x$ (both vanish at ± 1, independent). Isomorphism: $p + U \mapsto \big(p(1), p(-1)\big)$. Basis of the quotient: $1 + U$, $x + U$ (they map to the basis $(1, 1), (1, -1)$).

(b) No: $D(x^2 - 1) = 2x$, and $2x$ is not 0 at 1. The rule is not well defined: $(x^2 - 1) + U = 0 + U$, but the rule sends these to $2x + U$ and $0 + U$, which differ because $2x \notin U$.

(c) $p(1) = p(-1) = 0$ implies $p(1) = 0$, so $U \subseteq W$. By the third isomorphism theorem, $(\mathcal{P}_3(\mathbb{R})/U)/(W/U) \cong \mathcal{P}_3(\mathbb{R})/W$, and $\mathcal{P}_3(\mathbb{R})/W \cong \mathbb{R}$ via $p + W \mapsto p(1)$ (surjective, null space W). Dimension check: $2 - 1 = 1$.

Problem 6

(a) Over $\mathbb{C}$, m_T splits. Suppose some root λ is repeated: $m_T = (x-\lambda)^2 q$. Since $(x-\lambda)q$ has smaller degree than m_T, there is v with $(T-\lambda I)q(T)v \neq 0$. Put $u = q(T)v$. Then $(T-\lambda I)^2 u = m_T(T)v = 0$, so $u \in \operatorname{null}(T-\lambda I)^2 = \operatorname{null}(T-\lambda I)$, so $(T-\lambda I)u = 0$: contradiction. So m_T has no repeated root, and T is diagonalizable (Prop. 68).

(b) If T is diagonalizable, $T-\alpha I$ is diagonal in an eigenbasis, and a diagonal matrix and its square have the same zero diagonal entries, so the same null space.

(c) For α not an eigenvalue both null spaces are $\{0\}$. Example: $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\alpha = 0$: $\operatorname{null} N = \operatorname{span}(e_1)$, $\operatorname{null} N^2 = \mathbb{C}^2$.

Problem 7

(a) $m_{T^2} = \prod(y-\mu_i)$ with μ_i distinct and nonzero (T^2 invertible). Then $q(x) = \prod(x^2-\mu_i)$ satisfies $q(T) = m_{T^2}(T^2) = 0$, and its roots $\pm\sqrt{\mu_i}$ are distinct: $\sqrt{\mu_i} \neq -\sqrt{\mu_i}$ since $\mu_i \neq 0$, and different μ_i give different squares. So T is diagonalizable.

(b) $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: $N^2 = 0$ is diagonalizable, N is not.

(c) In an eigenbasis $T = \operatorname{diag}(\lambda_i)$; let $S = \operatorname{diag}(\sqrt{\lambda_i})$ for any complex square roots.

(d) Suppose $S^2 = N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Then $SN = S^3 = NS$, so S commutes with N, hence with $I+N$, so $S = \left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right)$ (T3(c)). Then $S^2 = \left(\begin{smallmatrix} a^2 & 2ab \\ 0 & a^2 \end{smallmatrix}\right) = N$ forces $a = 0$, and then $S^2 = 0 \neq N$. Contradiction.

Problem 8

(a) Induction on $\dim V$; $\dim V = 1$ is clear. By T3, S and T have a common eigenvector v. $U = \operatorname{span}(v)$ is invariant under both, and S/U, T/U commute: $(S/U)(T/U)(w+U) = STw + U = TSw + U$. By induction (applied to V/U, of smaller dimension, when it is nonzero), some basis $v_2+U, \ldots, v_n+U$ of V/U makes both quotient operators upper-triangular. Then $v, v_2, \ldots, v_n$ is a basis of V, and $Tv_j \in \operatorname{span}(v, v_2, \ldots, v_j)$ because $Tv_j + U \in \operatorname{span}(v_2+U, \ldots, v_j+U)$. Same for S.

(b) In that basis, $\mathcal{M}(S+T) = \mathcal{M}(S) + \mathcal{M}(T)$ and $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$ are upper-triangular with diagonals $\alpha_i + \beta_i$ and $\alpha_i\beta_i$, where α_i, β_i are the diagonal entries of $\mathcal{M}(S), \mathcal{M}(T)$, which are eigenvalues of S, T. The eigenvalues of an upper-triangular matrix are its diagonal entries.

(c) S and T are triangular with zero diagonal, so their only eigenvalue is 0. But $S+T = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has eigenvalues ± 1, and $ST = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ has eigenvalue 1. ($ST \neq TS$.)

Problem 9

(a) $U \cap W = \{p : p(0) = p(1) = 0\}$ has dimension 2 (basis $x^2 - x$, $x^3 - x$). $\dim(U+W) = 3+3-2 = 4$, so $U + W = \mathcal{P}_3(\mathbb{R})$. The isomorphism is $p + (U \cap W) \mapsto p + W$ for $p \in U$. Both spaces are 1-dimensional: $U/(U \cap W) \cong \mathbb{R}$ via $p + (U \cap W) \mapsto p(1)$, and $\mathcal{P}_3(\mathbb{R})/W \cong \mathbb{R}$ via $p + W \mapsto p(1)$. Basis: $x + (U \cap W)$, which maps to $x + W$ and then to 1.

(b) The map $W \to V/U$, $w \mapsto w + U$, is linear. Injective: $w + U = 0 + U$ means $w \in U \cap W = \{0\}$. Surjective: write $v = u + w$; then $v + U = w + U$. (Or: second isomorphism theorem, $(U+W)/U \cong W/(U \cap W) = W/\{0\} \cong W$.)

(c) Apply the first isomorphism theorem to $T|_U : U \to W$: its range is $T(U)$ and its null space is $U \cap \operatorname{null} T$.

Problem 10

(a) Extend to a basis $v_1, \ldots, v_n$ of V and take the first m functionals of the dual basis.

(b) "$\subseteq$" is clear. If $v \notin W$, the separating functional (Section 1, Annihilators) gives $\varphi \in W^0$ with $\varphi(v) = 1 \neq 0$.

(c) ($\Rightarrow$) clear. ($\Leftarrow$) If $W^0 \subseteq U^0$ and $u \in U$, then every $\varphi \in W^0$ is 0 at u, so $u \in W$ by (b).

Problem 11

(a) If U is invariant, $T|_U$ is diagonalizable (restrictions of diagonalizable operators are, Section 1.14), so U has a basis of eigenvectors of T. The eigenspaces of T are the three coordinate lines, so each of these eigenvectors is a multiple of some e_j. Hence U is the span of a subset of $\{e_1, e_2, e_3\}$: 8 invariant subspaces.

(b) $\{0\}$, $\operatorname{span}(e_1)$, $\operatorname{span}(e_1, e_2)$, $\mathbb{C}^3$. These are invariant since $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$. Conversely, a nonzero invariant U contains an eigenvector of $N|_U$, which lies in $\operatorname{null} N = \operatorname{span}(e_1)$, so $e_1 \in U$. If U also contains

$v = ae_1 + be_2 + ce_3$ with $c \neq 0$, then $N^2 v = ce_1$, $Nv = be_1 + ce_2$ give $e_2 \in U$, and then $e_3 \in U$: $U = \mathbb{C}^3$. Otherwise every vector of U has $c = 0$; if some has $b \neq 0$, then $U = \operatorname{span}(e_1, e_2)$, and if none does, $U = \operatorname{span}(e_1)$.

(c) Every subspace: $\{0\}$, $\mathbb{C}^2$, and every line (infinitely many).

Problem 12

(a) $(T - I)^2 = 0$, so every eigenvalue is a root of $(x-1)^2$, and 1 is an eigenvalue because $V \neq \{0\}$ (T1). m_T divides $(x-1)^2$, so $m_T = x - 1$ (then $T = I$, diagonal) or $m_T = (x-1)^2$ (repeated root, not diagonalizable, Prop. 68).

(b) $V \neq \{0\}$ and the polynomial splits over $\mathbb{R}$, so some factor is not injective (T1); distinct roots, so diagonalizable (T2).

(c) $T(T^2 - 2I) = -5I$, so $T^{-1} = -\frac{1}{5}(T^2 - 2I) = \frac{1}{5}(2I - T^2)$. Dividing $T^3 - 2T + 5I = 0$ by T^3: $I - 2T^{-2} + 5T^{-3} = 0$, so T^{-1} satisfies $y^3 - \frac{2}{5}y^2 + \frac{1}{5}$. It has degree 3; a smaller relation for T^{-1} would give one for T of degree < 3 (multiply by a power of T), so this is $m_{T^{-1}}$.

§6 True or false

1. True: $(v, w) \mapsto (w, v)$.
2. False: $\dim(V \times W) = \dim V + \dim W$ (Prop. 36).
3. False: three different lines in $\mathbb{R}^2$.
4. True: $0 \in v + U$ iff $-v \in U$ iff $v \in U$.
5. False: it means $v - w \in U$. Example: $U = \operatorname{span}(e_1)$, $v = e_1$, $w = 0$.
6. True: $\pi\colon V \to V/\{0\}$ is injective; and every $v + V = 0 + V$.
7. True: first isomorphism theorem with range $T = W$.
8. False: $U = \operatorname{span}(e_1)$, $W = \operatorname{span}(e_2)$ in $\mathbb{R}^2$; both quotients are 1-dimensional.
9. True: second isomorphism theorem.
10. True: given $S\colon V/U \to W$, take $T = S \circ \pi$.
11. True: Prop. 42 with $U_2 = \{0\}$; if $v - v' \in U \subseteq \operatorname{null} T$ then $Tv = Tv'$.
12. True: φ is surjective onto $\mathbb{F}$; rank–nullity.
13. True: if $\varphi_2 = 0$ then $\varphi_1 = 0$. Otherwise pick v with $\varphi_2(v) = 1$ and set $c = \varphi_1(v)$; $\varphi_1 - c\varphi_2$ is 0 on $\operatorname{null}\varphi_2$ and at v, which together span V.
14. False: T injective iff T' surjective. The inclusion $\mathbb{F} \to \mathbb{F}^2$ is injective and its dual is not.
15. False: it is $U^0 + W^0$. For two different lines U, W in $\mathbb{R}^2$, $(U \cap W)^0 = V'$ but $U^0 \cap W^0 = \{0\}$.
16. False: in $\mathbb{R}^2$ with $v_1 = e_1$, $v_2 = e_2$, $\varphi_1 = x$; with $v_1 = e_1$, $v_2 = (1, 1)$, $\varphi_1 = x - y$.
17. False: rotation R on $\mathbb{R}^2$.
18. False: $T = I$ has no eigenvalue 0.
19. False: e_1 and e_2 for $\operatorname{diag}(1, 2)$; their sum is not an eigenvector.
20. True. Let $Tv_i = \lambda_i v_i$ and $T(v_1 + v_2) = \mu(v_1 + v_2)$. If $\lambda_1 \neq \lambda_2$, then v_1, v_2 are independent, and $\lambda_1 v_1 + \lambda_2 v_2 = \mu v_1 + \mu v_2$ gives $\lambda_1 = \mu = \lambda_2$, a contradiction. So $\lambda_1 = \lambda_2 = \lambda$, and $T(v_1 + v_2) = \lambda(v_1 + v_2)$ with $v_1 + v_2 \neq 0$, so $\mu = \lambda$.
21. True: $T^2 v = \lambda^2 v$.
22. False: N and 0.
23. True: $T|_U$ is injective on the finite-dimensional U.
24. False: I.
25. True: distinct eigenvalues, as many as the dimension.
26. True: in an eigenbasis T^{-1} is diagonal with entries $1/\lambda_i$.
27. True: $m_{T|_U}$ divides m_T.
28. False: $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ (eigenvalues $1, 0$) and $T = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$; $S + T = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$.
29. True: $x^2 - 1 = (x-1)(x+1)$ has distinct real roots.
30. False: N. (True only for $T = 0$.)
31. False: rotation by 90° on $\mathbb{R}^2$ (no eigenvalue, so no first basis vector can be an eigenvector).
32. True: Prop. 67.
33. False: $T = I$, $p = (x-1)(x-2)$.
34. True: $m_T(0) = 0$ would make 0 an eigenvalue.

35. False: $\mathrm{diag}(1, 1, 2)$.
36. True. Let $v = w_1 + \cdots + w_n$ with w_i an eigenvector for λ_i (all different). If $\sum_{j<n} c_j T^j v = 0$, put $q = \sum_j c_j x^j$; then $0 = q(T)v = \sum_i q(\lambda_i) w_i$, so $q(\lambda_i) = 0$ for all n values λ_i. A polynomial of degree less than n with n roots is 0, so all $c_j = 0$. Hence $v, Tv, \ldots, T^{n-1}v$ is a basis.
37. True: $x \mapsto Ax$ is an operator on the odd-dimensional real space $\mathbb{R}^5$, so it has a real eigenvalue (Real operators (2), Section 1.14), and hence a real eigenvector.